MIL-A-83162(USAF)
AMENDMENT 3
15 January 1971

SUPERSEDING
AMENDMENT 2
7 January 1970

MILITARY SPECIFICATION

AMPLIFIER, ELECTRONIC CONTROL
SYNCHRO SIGNAL AM-6015/A

This amendment forms a part of Military Specification
MIL-A-83162(USAF) dated 31 July 1968.

Page 3, paragraph 3.2.3.4.3: Delete and substitute:

"3.2.3.4.3 Repeater excitation. The amplifier shall supply two modes of voltage excitation to the torque receivers. The primary mode shall be 26V, 400 Hz, at 25 watts maximum; this mode shall be supplied to pin B on the J-20 receptacle as specified in table I. The secondary mode shall be 86.5V, 400 Hz, at 10 watts maximum. When required, the secondary mode shall be obtainable, by internal wiring modifications, on pin B of the J-20 receptacle.

NOTE: All torque receivers driven from this amplifier should be excited from either the 115V, 400 Hz excitation to the AM-6015/A amplifier or by either of the two modes specified in 3.2.3.4.3."

Page 3, paragraph 3.4.3: Delete and substitute:

"3.4.3 Accuracy. The amplifier shall not introduce more than 0.15 degree of error into the torque receiver load specified in 3.4.2."

Page 6, paragraph 3.4.4: Delete and substitute:

"3.4.4 Multiple load isolation. The displacement of one of the multiple loads 20 degrees from the transmitter output shall not cause any other receiver load or transmitter to vary more than 0.25 degree."

Page 7, paragraph 4.5.1: Add:

"Impedance 4.6.12"

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Page 8, paragraph 4.5.2.2, line 4: Delete and substitute:

"Post-overload accuracy (The amplifier shall operate with a short for 10 minutes) 4.6.7.1"

Page 9, paragraph 4.6.4, line 3: Delete "0.5 percent" and substitute "0.2 percent."

Page 9, paragraph 4.6.5, line 5: Delete "0.2 degree" and substitute "0.15 degree."

Page 9, paragraph 4.6.6, line 4: Delete "0.3 degree" and substitute "0.25 degree."

Page 9, paragraph 4.6.7: Delete and substitute:

"4.6.7 Overload tests

"4.6.7.1 Post-overload accuracy test. The amplifier shall be connected with a data transmitter (or equivalent) input. Both outputs shall be shorted to the ground and the amplifier allowed to operate for five minutes. Immediately following, the amplifier shall be connected as specified in 4.6.5 and rechecked for accuracy as specified in 4.6.5, except the measurement shall be every 90 degrees. The same accuracy tolerance shall apply.

"4.6.7.2 Overload monitor. The amplifier shall be connected with a load of four torque receivers, type 26V-11-TR4C (or equivalent). One of the torque receivers shall be displaced 90 degrees. During the overload condition, the remaining three torque receivers shall oscillate between the heading input and the heading of the displaced torque receiver, indicating a torque receiver load malfunction."

Page 9, paragraph 4.6.8: Delete and substitute:

"4.6.8 Repeater excitation voltage test. The primary receiver excitation voltage of 26V ±1V, 400 Hz, at 26 watts maximum, shall be measured across a 26-ohm 50-watt resistor. The secondary receiver excitation voltage of 86.5V ±3.5V, 400 Hz, at 10 watts maximum, shall be measured across an 1000-ohm 10-watt resistor.



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Page 9, paragraph 4.6.9.1, subparagraph a: Delete and substitute:

"a. Temperature and altitude: The tolerance specified in 4.6.5 shall be 0.25 degree; the tolerance specified in 4.6.6 shall be 0.30 degree."

Page 10: Add:

"4.6.12 Impedance tests. The supplier shall conduct tests to determine compliance with the requirements specified in 3.4.1.1 and 3.4.1.2 herein."

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